Practical No.: 07

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COURSE: AI&DS
CLASS: BE
SUB: Computer Laboratory-IV

Title: Data Visualization from Extraction Transformation and Loading (ETL) Process.

0.1 ETL Assignment Implementation

0.1.1 Extract Data

[15]:
```
import pandas as pd

# Extract data from CSV
df = pd.read_csv("sales_and_customer_insights.csv")

# Display first 5 rows
df.head()
```

[15]:

	Customer_ID	Product_ID	Transaction_ID	Purchase_Frequency	\
0	CUST_9HOS83	PROD_IK97D1	TRANS_II1DZG	17	
1	CUST_AJU17N	PROD_UNN7KP	TRANS_9HJF7I	10	
2	CUST_11XNYF	PROD_0XEW2W	TRANS_OT96OM	3	
3	CUST_IGH8G3	PROD_3IIAJN	TRANS_45V00G	12	
4	CUST_OK6PUM	PROD_VMIWD2	TRANS_ZAK760	18	

	Average_Order_Value	Most_Frequent_Category	Time_Between_Purchases	\
0	172.57	Electronics	45	
1	64.89	Clothing	6	
2	120.38	Sports	23	
3	70.34	Clothing	5	
4	42.39	Electronics	10	

	Region	Churn_Probability	Lifetime_Value	Launch_Date	\
0	South America	0.98	952.81	2020-03-14	
1	South America	0.66	5427.51	2022-10-15	
2	Asia	0.60	3994.80	2021-11-30	
3	North America	0.78	4880.01	2022-03-20	
4	North America	0.52	8557.70	2022-11-09	

	Peak_Sales_Date	Season	Preferred_Purchase_Times	Retention_Strategy
0	2023-09-11	Winter	Afternoon	Loyalty Program
1	2023-01-02	Spring	Afternoon	Discount
2	2023-04-06	Winter	Evening	Loyalty Program
3	2023-03-23	Spring	Evening	Discount
4	2023-10-28	Spring	Morning	Discount

df.info() df.describe()

0.1.2 Transform Data (Data Cleaning & Preparation)

[16]:
```
df.isnull().sum()
```

[16] :

Customer_ID	0
Product_ID	0
Transaction_ID	0
Purchase_Frequency	0
Average_Order_Value	0
Most_Frequent_Category	0
Time_Between_Purchases	0
Region	0
Churn_Probability	0
Lifetime_Value	0
Launch_Date	0
Peak_Sales_Date	0
Season	0
Preferred_Purchase_Times	0
Retention_Strategy	0

dtype: int64

[17] :
```
df = df.dropna()
```

[18] :
```
df['Launch_Date'] = pd.to_datetime(df['Launch_Date'])
df['Peak_Sales_Date'] = pd.to_datetime(df['Peak_Sales_Date'])
```

[19] :
```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[['Purchase_Frequency','Average_Order_Value','Lifetime_Value']] = scaler.
 fit_transform(
    df[['Purchase_Frequency','Average_Order_Value','Lifetime_Value']]
)
```

0.1.3 Load Data (Save Clean Dataset)

[20] :
```
df.to_csv("cleaned_sales_data.csv", index=False)

print("Cleaned data saved successfully")
```

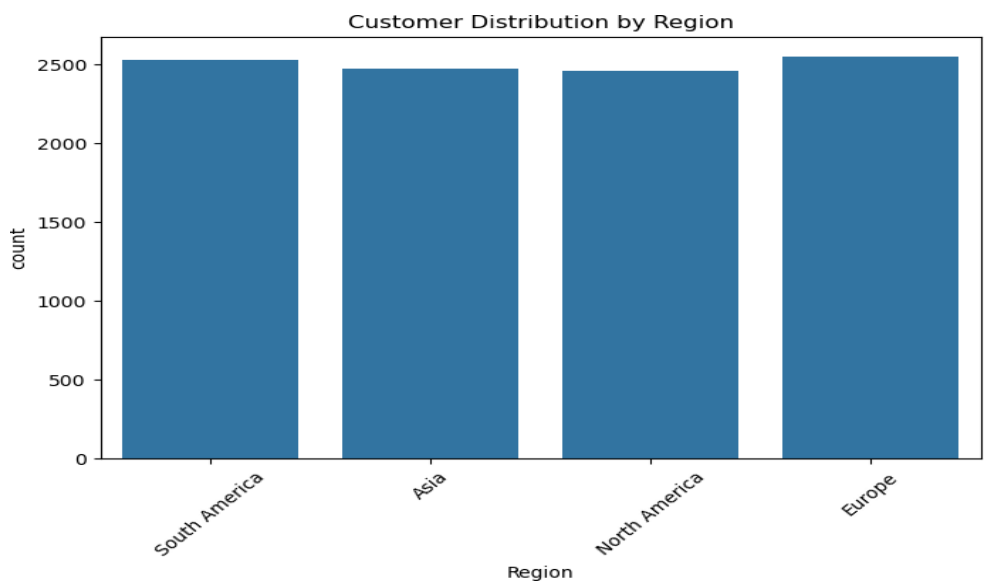
Cleaned data saved successfully

0.1.4 Data Visualization

[21] :
```
import matplotlib.pyplot as plt
import seaborn as sns
```

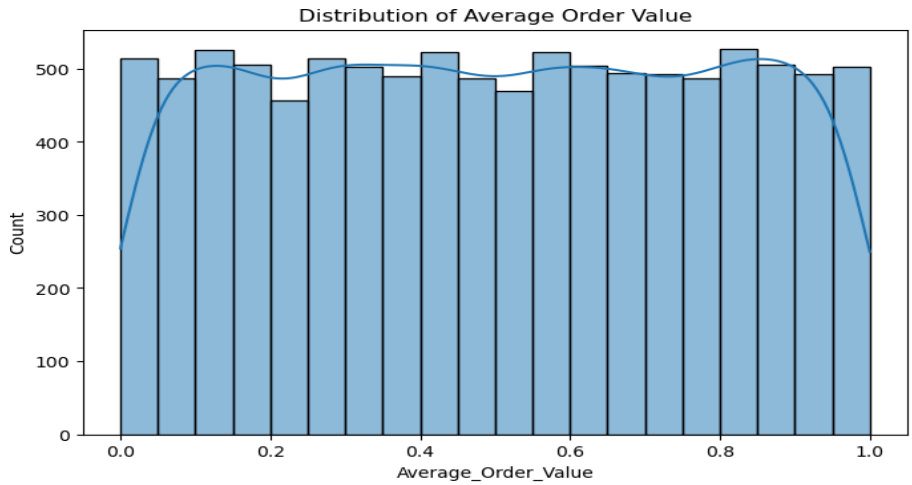
0.1.5 Visualization 1 – Bar Chart (Region Distribution)

[22] :
```
plt.figure(figsize=(8,5))
sns.countplot(x='Region', data=df)
plt.title("Customer Distribution by Region")
plt.xticks(rotation=45)
plt.show()
```



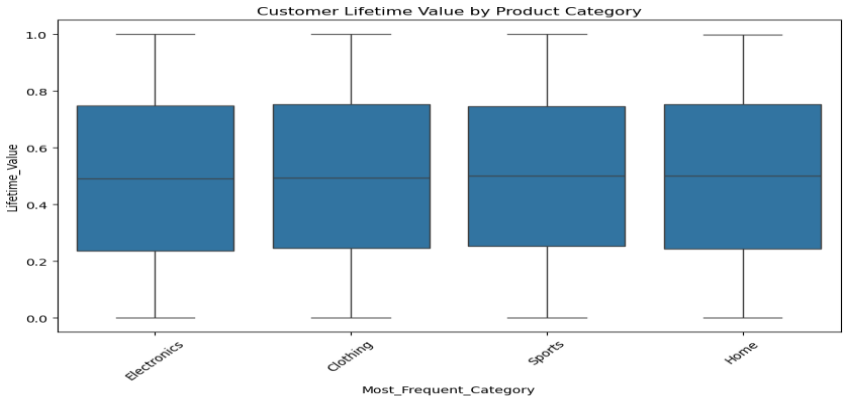
0.1.6 Visualization 2 – Histogram (Average Order Value)

[23] :
```
plt.figure(figsize=(8,5))
sns.histplot(df['Average_Order_Value'], bins=20, kde=True)
plt.title("Distribution of Average Order Value")
plt.show()
```



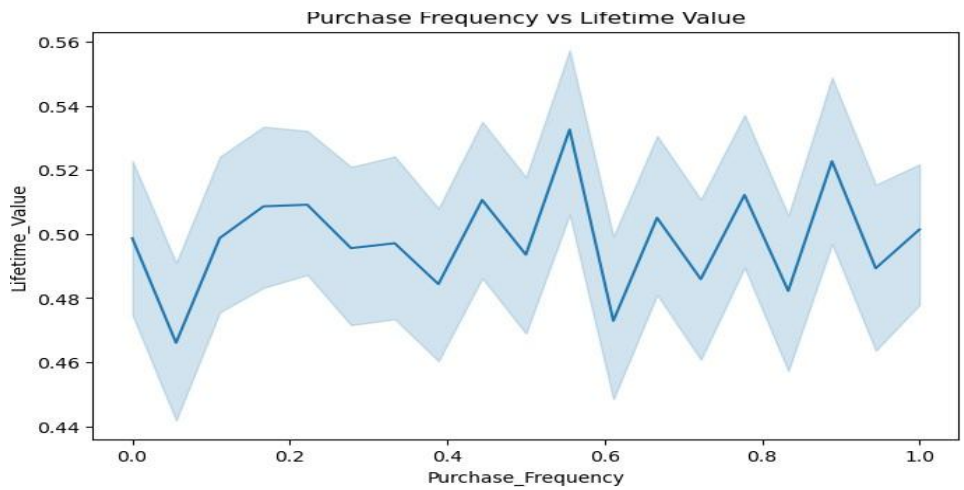
0.1.7 Visualization 3 – Box Plot (Lifetime Value by Category)

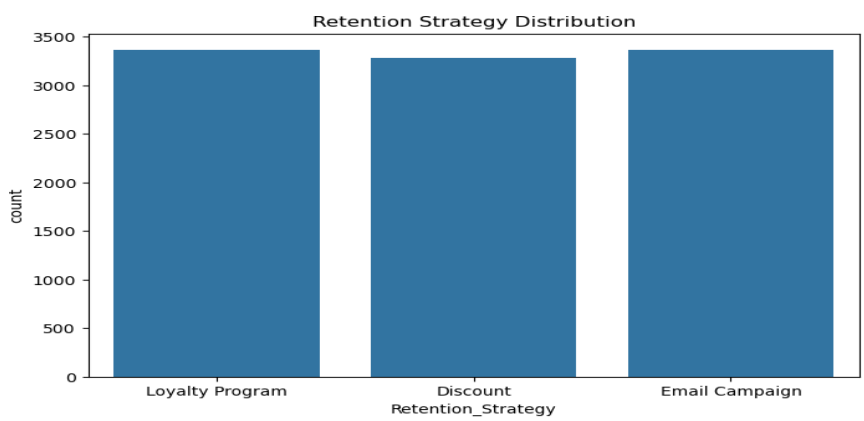
[24] :
```
plt.figure(figsize=(10,6))
sns.boxplot(x='Most_Frequent_Category', y='Lifetime_Value', data=df)
plt.title("Customer Lifetime Value by Product Category")
plt.xticks(rotation=45)
plt.show()
```



0.1.8 Visualization 4 – Line Plot (Purchase Frequency vs Lifetime Value)

[25] :
```
plt.figure(figsize=(8,5))
sns.lineplot(x='Purchase_Frequency', y='Lifetime_Value', data=df)
plt.title("Purchase Frequency vs Lifetime Value")
plt.show()
```



0.1.9 Visualization 5 – Bar Chart (Retention Strategy)

[26] :
```
plt.figure(figsize=(8,5))
sns.countplot(x='Retention_Strategy', data=df)
plt.title("Retention Strategy Distribution")
plt.show()
```